

Assignment 3 - Harris Shibli (28988)

Qs1. T = Temperature sensor (1 if $\geq 30^{\circ}\text{C}$) | P = Pressure sensor (1 if $\geq 50\text{ psi}$)

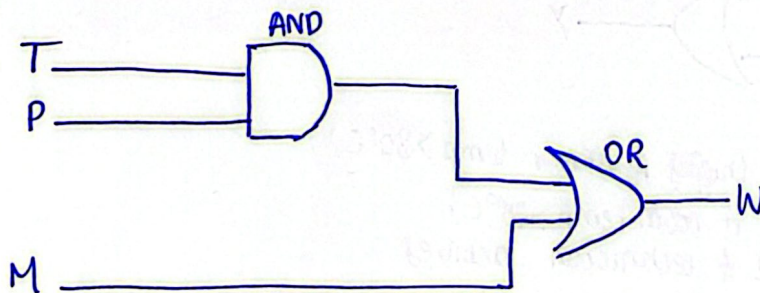
M = Master alarm switch (1 if activated)

Truth Table

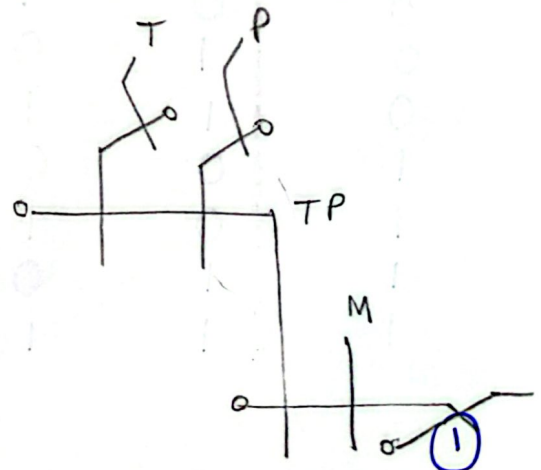
T	P	M	W
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$W = M + (T \cdot P)$$

* The warning light turns on if the master switch is on, OR if both temperature and pressure exceed the desired limit.



Qs2. ~~L = Water level sensor (1 if $> 10\text{ cm}$) | R = Rain forecast sensor (1 if there is heavy rain expected) | B = Backup battery sensor (1 if fully charged)~~

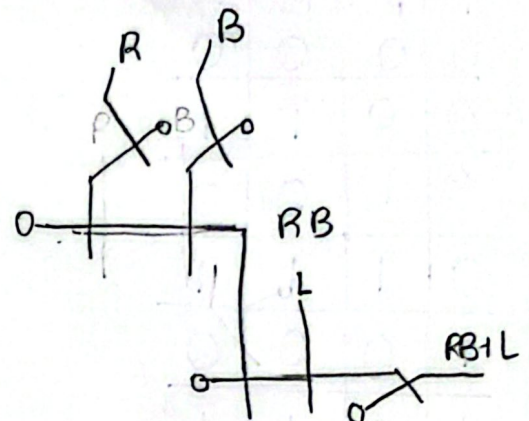


Q2A L = water level sensor (1 if > 100cm)

R = Rain forecast sensor (1 if heavy rain expected)

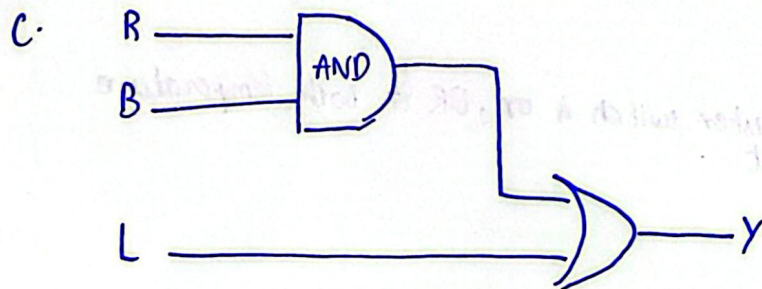
B = Backup battery sensor (1 if fully charged)

L	R	B	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



B. Boolean Expression $Y = L + (R \cdot B)$

This means that the pump will activate whenever the water level is high, OR when rain is expected and backup battery is fully charged.



Q3A S = Primary thermostat (1 if server temp > 80°C)

A = Ambient room sensor (1 if room temp > 25°C)

M = Maintenance switch (1 if technician activates)

S	A	M	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

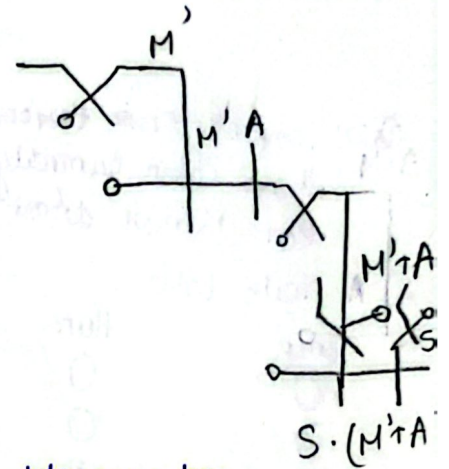
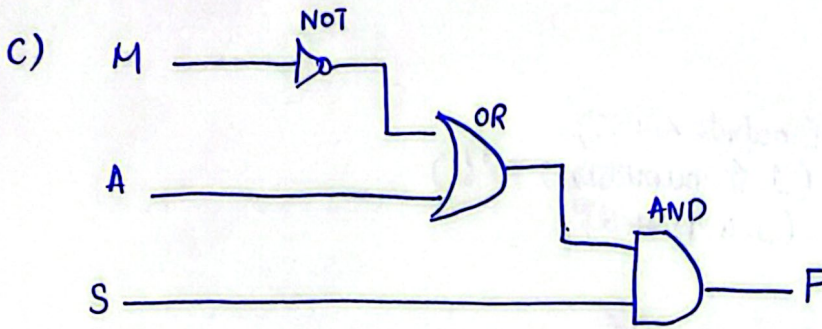


B) $F = S \cdot (M' + A)$

③

this can also be expressed as $F = (S \cdot M') + (S \cdot A)$

This is because the fan runs when the server is overheating and also when no technician is present, and also when server is overheating and the room is hot. Also when a technician is present.



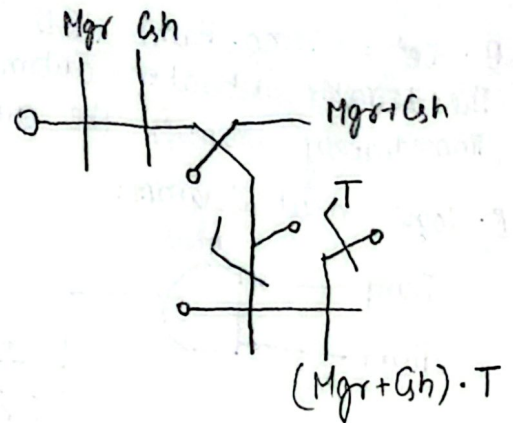
Qs 4. A.) $T = \text{Time lock (1 high at 9am-5pm)}$

Mgr = Manager's key (1 high if inserted and turned) - Manager's key.

Csh = Cashier's key (1 if inserted and turned) - this refers to Cashier's key

Truth Table

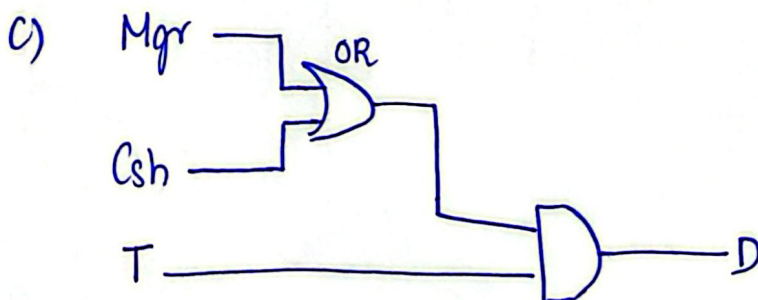
T	Mgr	Csh	D
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



B) Boolean Expression

$D = T \cdot (Mgr + Csh)$

This is translated to the fact that the vault unlocks only during business hours AND when atleast one authorized key is turned. Outside the business hours the vault doesn't unlock.



- Q5. Temp = Exterior thermometer (1 if outside $\leq 4^{\circ}\text{C}$)
Hum = Cabin humidity sensor (1 if humidity $> 70\%$)
Btn = Manual defrost button (1 if pressed)

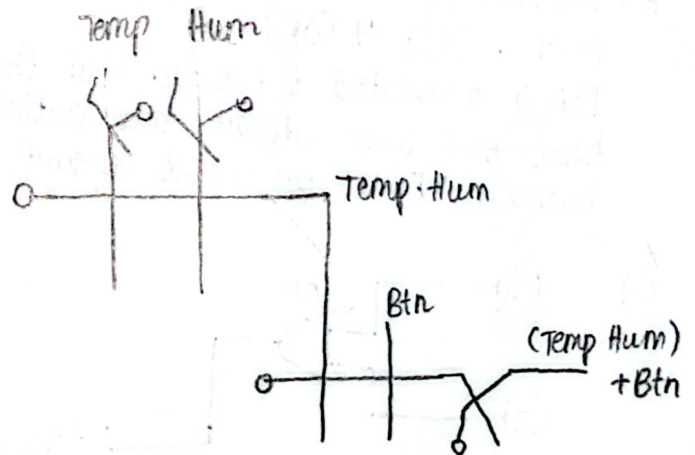
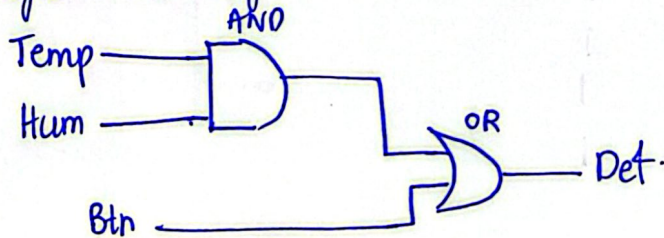
A Truth table

Temp	Hum	Btn	Def
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

B. $\text{Def} = (\text{Temp} \cdot \text{Hum}) + \text{Btn}$

The defroster activates automatically when it is both cold and humid, or immediately whenever the manual button is pressed

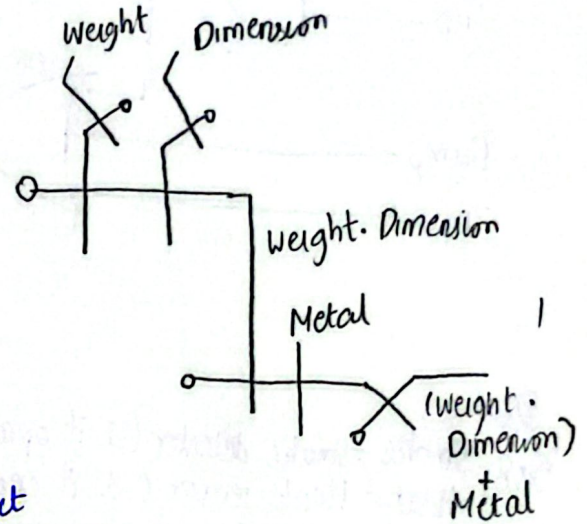
C. logic circuit diagram:



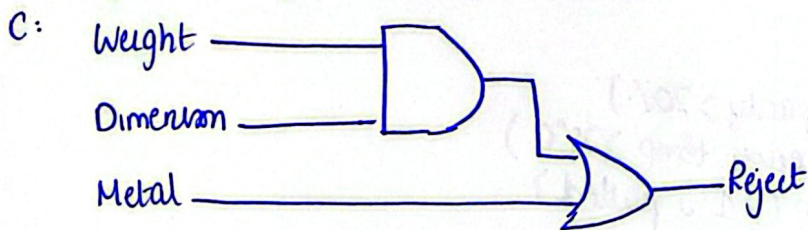
Qs6. Factory Conveyor Reject Mechanism

Task A:-

Weight	Dimension	Metal	Reject
0	1	0	0
1	0	0	0
0	0	0	0
1	1	0	1
0	1	1	1
1	0	1	1
0	0	1	1



B: $\text{Reject} = (\text{Weight} \cdot \text{Dimension}) + \text{Metal}$



Qs7 A. Trip = laser tripwire (1 if broken/person present)
 Wt = weight limit sensor (1 if > 1000 kgs)
 Bump = Mechanical bumper (1 if it strikes an object)

Truth Table

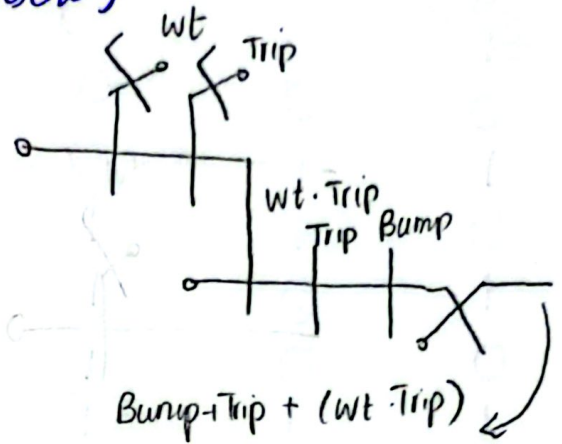
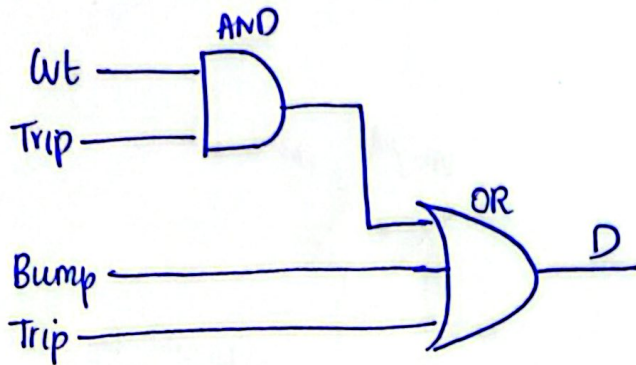
Trip	Wt	Bump	D
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

B. Boolean Expression

$$D = \text{Bump} + \text{Trip} + (\text{Wt} \cdot \text{Trip})$$

Simplifies to: $D = \text{Bump} + \text{Trip}$, since Trip already covers the $\text{Wt} \cdot \text{Trip}$ case.
The door reopens if the bumper strikes something, or if the tripwire is broken (with or without an overweight condition present.)

C:



Q8.

Smoke = smoke detector (1 if opacity > 20%)
Heat = Heat sensor (1 if ceiling temp > 75°C)
Pull = Manual pull station (1 is pulled)

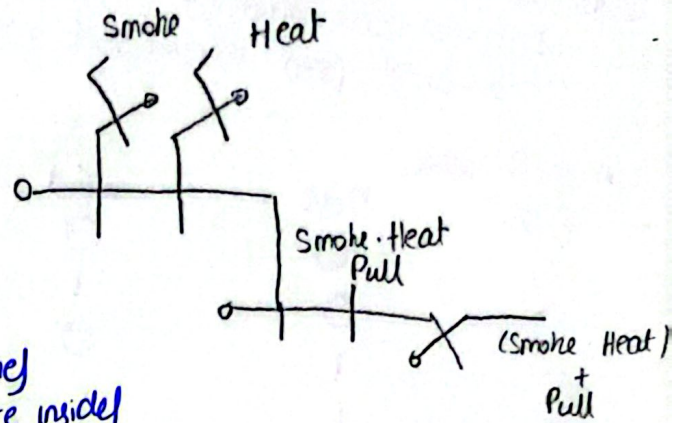
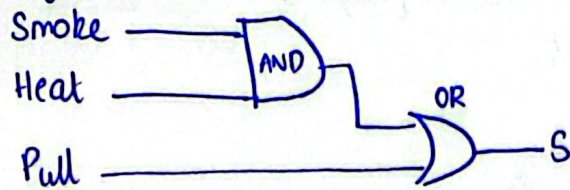
A.	Smoke	Heat	Pull	S
	0	0	0	0
	0	0	1	1
	0	1	0	0
	0	1	1	1
	1	0	0	0
	1	0	1	1
	1	1	0	1
	1	1	1	1

B. Boolean Expression:

$$S = (\text{Smoke} \cdot \text{Heat}) + \text{Pull}$$

The sprinkler trigger automatically only when both smoke and heat thresholds are crossed together or immediately it manually pulled.

C: Logic Circuit Diagram



Q9.

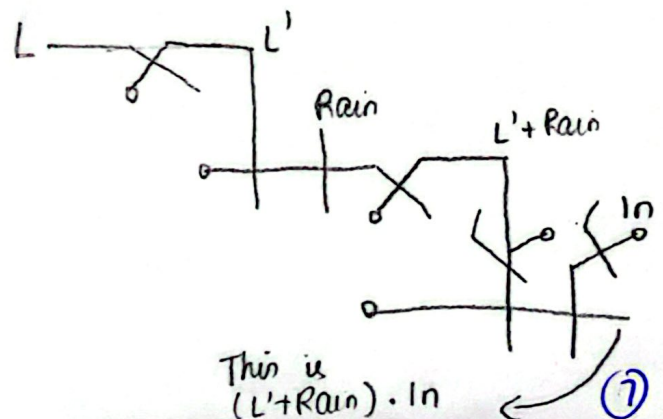
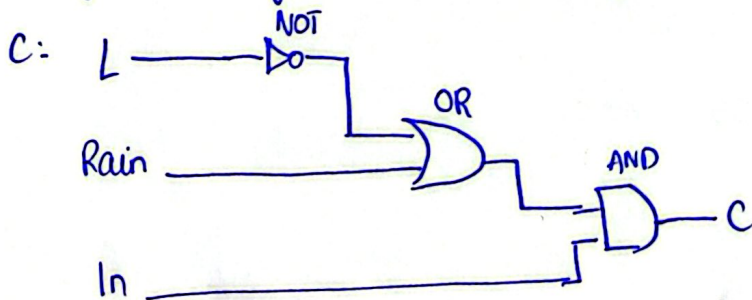
L = light sensor (1 if > 500 lux/daytime)
 In = Motion sensor (1 if all chickens are inside)
 Rain = Rain sensor (1 if currently raining)

A: Truth Table

L	In	Rain	C
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

B: Boolean Expression

$C = In \cdot (L' + Rain)$ The door closes at night as long as chickens are inside, and also ~~also~~ closes during the day if it starts raining - but only once every chicken is safely inside



Q10. Bat = Battery sensor (1 if $\leq 15\%$ charged)
 Dist = Distance sensor (1 if $> 2\text{km}$ from launch pad)
 Wind = Wind speed sensor (1 if $> 20\text{km/h}$)

A: Truth Table

Bat	Dist	Wind	Sv
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

B: $Sv = Bat \cdot (Dist + Wind)$

Battery saver mode is engaged whenever the battery is low AND either the drone is far from home or wind conditions are strong.

